Conditional Mutual Information for Screening Point-in-Time Macroeconomic Indicators

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Prior Work in Information-Theoretic Finance

Why Not Correlation

Correlation and linear regression, the default tools for testing whether one series predicts another, test only linear dependence, or, in the rank-correlation case, monotonic dependence. Zero correlation does not imply independence. For instance, if $Y = g(X)$ for a symmetric nonlinear g, for instance $Y = X^2$ with X symmetric about zero, $\mathrm{Corr}(X, Y) = 0$ even though Y is a deterministic function of X. This is not a pathological special case for financial data. The relationship between a macroeconomic indicator and subsequent returns is commonly regime-dependent or threshold-like (e.g. credit growth below versus above a rate consistent with sustainable expansion, correlation estimated over a sample spanning both regimes averaging the two effects into a near-zero, and potentially misleading, statistic). A regression-based screen also requires specifying a functional form before estimation, so an indicator whose true relationship to returns is quadratic, threshold-based, or conditional on another variable can receive a small or statistically insignificant coefficient regardless of how much of the variation in returns it actually explains. Mutual information does not have this limitation, since it is defined directly from the joint distribution of X and Y and is zero if and only if X and Y are independent, with no assumption on the functional form of the dependence. This is the property that motivates its use as a market-efficiency and dependence measure, in place of correlation-based tests.

Entropy and Efficiency Testing

Applying this idea to markets required first replacing linear-predictability tests with an entropy-based alternative. Liu et al. [2021] do so directly, testing market efficiency across global equity indices by estimating the Shannon entropy, instead of linear autocorrelation, of return distributions from histograms, on the grounds that entropy captures any departure from an unpredictable, maximum-entropy outcome distribution. Their estimator depends on a chosen bin width, and raw entropy is unbounded and scaled by the underlying return volatility, so a value from one market or period is not directly comparable to another without further normalisation. Patra and Mohapatra [2022] extend the approach to a broader set of markets and confirm the pattern holds outside a single index, but inherit the same histogram-based estimator, and therefore the same comparability problem, since the underlying quantity is still unbounded raw entropy.

Normalised Mutual Information and Transfer Entropy

Noguer i Alonso and Zoonekynd [2024] address the comparability problem directly by introducing normalised mutual information, $I(X; Y)/\sqrt{H(X)H(Y)}$, bounded in $[0, 1]$ by construction and therefore comparable across assets and periods regardless of differences in volatility, applying it across a cross-section of US stocks. Noguer i Alonso [2025], the paper this project extends, resolves two further gaps in that work. It replaces histogram-based estimation, which is sensitive to bin-width choice and does not extend naturally to continuous data, with the k-nearest-neighbour estimator of Kozachenko and Leonenko [1987], which requires no binning. It also introduces transfer entropy, since normalised mutual information is symmetric and cannot indicate which of two series is driving the dependence. Writing $X_t^{(p)} = (X_t, \ldots, X_{t-p+1})$ for the past p values of a series X (lag orders are written p, q in place of the more common k, ℓ, to keep k free for the nearest-neighbour count), transfer entropy is

$$T_{X\to Y} = I\big(X_t^{(p)};\, Y_{t+1} \mid Y_t^{(q)}\big),$$

the amount by which X's recent past reduces uncertainty about Y_{t+1} beyond what Y's own recent past, $Y_t^{(q)}$, already provides; it is zero when Y_{t+1} is independent of $X_t^{(p)}$ given $Y_t^{(q)}$, and reduces to ordinary Granger causality under a linear Gaussian model. Noguer i Alonso [2025] proves the relevant boundedness and consistency properties formally and applies the resulting toolkit, entropy, KL divergence, normalised mutual information, and transfer entropy, to twenty-five years of S&P 500 returns.

Conditional Mutual Information

Every measure in this sequence, from Liu et al. [2021] through Noguer i Alonso [2025], remains univariate. Normalised mutual information and transfer entropy are computed between a return series and its own history, or between two return series, and determine whether a series is predictable at all, not whether a specific external covariate remains informative once other available information is already accounted for. Transfer entropy in particular conditions only on the target's own past, $Y_t^{(q)}$; it says nothing about whether X remains informative once other indicators, not just Y's own history, are already known. The tool that generalises transfer entropy to an arbitrary conditioning set is conditional mutual information:

Definition 1 (Conditional mutual information)**.**

$$I(X;Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z) = \mathbb{E}_{X,Y,Z}\left[\log \frac{p(X,Y \mid Z)}{p(X \mid Z)\, p(Y \mid Z)}\right].$$

Equivalently, $I(X;Y \mid Z) = 0$ if and only if $X \perp Y \mid Z$; transfer entropy is recovered as the special case $Z = Y_t^{(q)}$.

$I(X;Y \mid Z)$ is the object relevant to indicator screening, namely whether X reduces uncertainty about Y beyond what Z already provides, for any Z, rather than whether X and Y are related unconditionally or beyond only Y's own history. This generalisation is developed outside the finance literature, in a strand of work concerned in turn with estimating $I(X;Y \mid Z)$ itself, testing whether an estimated value can be trusted, and making the resulting procedure usable at scale. Frenzel and Pompe [2007] take the first step, extending the Kozachenko-Leonenko estimator from the unconditional to the conditional case; Witter and Houghton [2024] strengthen this with an analytic bias correction and a proof of consistency under general regularity conditions, so the estimator's behaviour as sample size grows is well understood. An estimator alone is not enough, since a nonzero reading can arise from not dependence but sampling noise, which is the problem Runge [2018] addresses with a local permutation test for whether an estimated value should be believed. Runge [2019] assembles the estimator and the significance test into a full causal-discovery procedure, PCMCI, that also prunes candidate variables with an inexpensive unconditional test before running the expensive conditional test on what remains, because testing every variable pair directly does not scale. That the scaling concern is not hypothetical is confirmed directly by Causal Forecasting [2024], who report that running unpruned conditional independence testing on financial data, 136 predictors over roughly 200 time steps, took approximately two weeks of computation.

Remaining Gaps

No application of conditional mutual information or PCMCI-style testing to financial data addresses the two constraints that determine whether the extension above is usable at the scale of a full indicator universe. These are 1) that estimator accuracy degrades in the dimension of the conditioning set, as shown in the Estimator section below, and 2) that unpruned conditional testing is computationally prohibitive at that scale, as the runtime in Causal Forecasting [2024] demonstrates for a predictor set roughly two orders of magnitude smaller than JPMaQS's. JPMaQS's own cross-sectional structure offers a partial answer to the first of these constraints that has not been used for this purpose: the same metric is observed across twenty to fifty currency areas, and pooling the estimator's neighbour search across countries increases the effective sample size without enlarging the conditioning set, subject to the poolability assumption specified in the Panel Pooling section below. Neither constraint has been addressed for financial data specifically, and no paper identified above combines a conditional information measure, a panel structure exploited for sample size, and an explicit accounting of computational cost in one setting.

Problem

An indicator is useful for forecasting returns if it still carries information about where returns are headed once everything already available, price history and other indicators, is accounted for. Define r_t for the log return over day t, $h \geq 1$ for a forecast horizon so that r_{t+h} is realised h days after t, X_t for a candidate macroeconomic indicator, and $\mathcal{I}_t$ for the information available at t, assuming lagged returns and other indicators have already been observed. An indicator X_t is useful for forecasting r_{t+h} if $I(X_t; r_{t+h} \mid \mathcal{I}_t) > 0$. Linear regression tests only whether X_t has nonzero partial correlation with r_{t+h} given $\mathcal{I}_t$, which is necessary but not sufficient for $I(X_t; r_{t+h} \mid \mathcal{I}_t) > 0$ under nonlinear dependence. This project's methods estimate $I(X_t; r_{t+h} \mid \mathcal{I}_t)$ directly, using point-in-time macroeconomic indicators from the J.P. Morgan Macrosynergy Quantamental System (JPMaQS).

Estimator

Estimating $I(X; Y \mid Z)$ from the definition above requires estimating $H(Y \mid Z)$ and $H(Y \mid X, Z)$, and therefore the underlying joint and conditional densities. The direct approach, fitting a histogram or kernel density estimate and substituting it into the entropy formula, requires choosing a bin width or bandwidth in advance. This is the same problem already noted for the histogram-based entropy estimators of Liu et al. [2021] above, but in this case, it worsens as the number of conditioning variables grows, since a fixed bin width leaves most cells empty once the space being binned has more than one or two dimensions. The k-nearest-neighbour approach avoids fixing a bandwidth globally. For each data point, it considers only how far away that point's k-th nearest neighbour is. Specifically, a short distance means the data is locally dense there, so the local density is high and the local contribution to entropy is low, while a long distance means the data is locally sparse, so the local contribution to entropy is high. Averaging a function of these neighbour distances over all points gives an entropy estimate whose effective bandwidth adapts to the data itself.

Definition 2 (Frenzel–Pompe estimator, Frenzel and Pompe, 2007)**.** For N samples $\{(x_i, y_i, z_i)\}_{i=1}^N$, let ϵ_i be the distance from point i to its k-th nearest neighbour in the joint (x, y, z) space under the max norm, and let $n_z(i)$, $n_{xz}(i)$, $n_{yz}(i)$ be the number of points within distance ϵ_i of point i in the marginal z, (x, z), and (y, z) subspaces respectively. Then

$$\hat{I}(X; Y \mid Z) = \psi(k) + \frac{1}{N} \sum_{i=1}^{N} \Big[\psi\big(n_z(i) + 1\big) - \psi\big(n_{xz}(i) + 1\big) - \psi\big(n_{yz}(i) + 1\big) \Big],$$

where ψ is the digamma function.

The counts $n_{xz}(i)$ and $n_{yz}(i)$ measure how crowded point i's neighbourhood is once Y or X are respectively dropped from consideration. If X and Y are conditionally independent given Z, these marginal counts behave, on average, just as the joint neighbour count k and the marginal count $n_z(i)$ would predict under independence, and the bracketed digamma term averages to approximately zero across points. Real conditional dependence manifests as a recognisable difference between the joint and marginal neighbour counts, which the digamma differences convert into an entropy-scale quantity. The estimator only counts neighbours within the locally adaptive radius ϵ_i, which is what allows it to avoid a fixed bandwidth. Consistency under mild regularity conditions is established in Frenzel and Pompe [2007] and Witter and Houghton [2024].

Proposition 1 (Bias–variance rate)**.** *For conditioning-set dimension d and sample size N, the estimator's bias scales as $(k/N)^{2/d}$ and its variance as $1/k$, giving a mean-squared-error-optimal $k^\star \propto N^{4/(d+4)}$. The corresponding neighbour radius, $\epsilon_{k^\star} \sim (k^\star/N)^{1/d} \propto N^{-1/(d+4)}$, is the same rate that governs optimal bandwidth selection in kernel density estimation [Silverman, 1986].*

A larger k averages over a larger neighbourhood, which lowers variance but assumes local density is constant over a region where it may not be, raising bias; a smaller k accounts for local structure more closely, but since variance scales as $1/k$, fewer neighbours contributing to each point's local count increases that variance. As d increases, the optimal-rate exponent $4/(d+4)$ shrinks towards zero, so the optimal k grows only slowly with N and the estimator's achievable accuracy at fixed N degrades; in practice it is unreliable once $\dim(Z)$ exceeds three or four at the sample sizes available from a single country's daily return history ($N \sim 10^3$–10^4).

Panel Pooling

JPMaQS records the same economic metric, such as credit growth, across many countries at once, so rather than estimating the relationship separately in each country with too few points, the countries' histories can be treated as additional draws from a common underlying distribution. The constraint above concerns N relative to d, not N alone, and since $k^\star \propto N^{4/(d+4)}$, raising N while holding d fixed raises the achievable $k^\star$ and lowers the estimator's mean-squared error, even though d itself has not changed. JPMaQS indicators are observed across $C \approx 20$ to 50 currency areas for a given concept, and pooling the k-NN neighbour search across this cross-section increases the effective N available to the estimator at fixed d.

Definition 3 (Pooled estimator)**.** For countries $c = 1, \dots, C$, let $\tilde{x}_t^c$, $\tilde{y}_t^c$, $\tilde{z}_t^c$ denote indicator, return, and conditioning-set values standardised within country (subtracting the country-specific mean and dividing by the country-specific standard deviation, to remove level and scale differences across currency areas). The pooled estimator $\hat{I}^{\text{pool}}(X; Y \mid Z)$ applies the Frenzel–Pompe estimator to the union $\bigcup_c \{(\tilde{x}_t^c, \tilde{y}_t^c, \tilde{z}_t^c)\}_t$.

Standardisation is necessary before pooling since the same nominal indicator level (for example, a given percentage credit growth rate) need not correspond to the same economic state across countries with different baseline levels and volatility.

Pooling is only valid if $p(r_{t+h} \mid X_t, \mathcal{I}_t)$ is the same function across countries c. If it is not, the pooled estimate is not merely noisier than a country-specific one, it is biased, since it then describes some average relationship that may hold for no single country precisely. This has to be checked before pooling, so a natural diagnostic compares what pooling everything together gives against what the countries give separately:

Definition 4 (Poolability statistic)**.** Let $\hat{I}^{(c)}(X; Y \mid Z)$ denote the Frenzel–Pompe estimate computed from country c's observations alone. Define

$$\Lambda = \hat{I}^{\text{pool}}(X; Y \mid Z) - \frac{1}{C} \sum_{c=1}^{C} \hat{I}^{(c)}(X; Y \mid Z).$$

Under poolability, $\hat{I}^{\text{pool}}$ and every $\hat{I}^{(c)}$ are estimating the same population quantity $I(X; Y \mid Z)$, only from samples of different sizes, so all of them, and hence their average, should agree up to estimation noise, making $\Lambda \approx 0$. If instead the true relationship differs by country, $\hat{I}^{\text{pool}}$ is effectively estimating a sample-size-weighted mixture of the country-specific relationships rather than their simple average, and there is no reason this mixture should equal the unweighted average of the country-specific values; Λ then reflects that mismatch rather than pure sampling noise. To judge how large is large enough to matter, I construct a null distribution for Λ by permuting country labels across the panel while preserving each observation's within-country time index. This scrambles which country each observation is attributed to while leaving the time-series structure within each country intact, so any resulting pattern in Λ reflects the sampling process and avoids confounding with cross-country structure. Recomputing Λ under B such permutations and comparing the observed value to this distribution gives a test of poolability. Where the test rejects poolability at a chosen level, I use partial pooling instead of pooling outright,

$$\hat{I}_c^{\text{shrink}} = w_c \, \hat{I}^{(c)} + (1 - w_c) \, \hat{I}^{\text{pool}}, \qquad w_c = \frac{N_c}{N_c + \lambda},$$

an empirical-Bayes shrinkage rule of the form used in hierarchical panel models [Efron and Morris, 1975], where a country with abundant data of its own is trusted mostly on its own estimate, while a country with little data leans more heavily on the pooled estimate, with λ chosen by cross-validation. This differs from Macrosynergy's `CategoryRelations` treatment of common global factors, which adjusts standard errors for correlated residuals across countries under a shared coefficient, a weaker assumption than a shared conditional distribution.

Significance Testing

A nonzero sample value of $\hat{I}(X;Y \mid Z)$ does not by itself establish that X and Y are dependent given Z. The Frenzel–Pompe estimator is a finite-sample statistic and will generally return a positive value even when the true conditional mutual information is zero, purely from sampling noise in the neighbour counts. Determining whether an estimated value is distinguishable from what noise alone would produce requires comparing it to a null distribution through a significance test.

Definition 5 (Leakage-safe local permutation test, adapted from Runge, 2018)**.** For each t, define the permutation neighbourhood

$$\mathcal{N}(t) = \{t' \leq t : z_{t'} \text{ is among the } m \text{ nearest neighbours of } z_t \text{ in } Z\text{-space}\},$$

restricted to $t' \leq t$ so that no permutation uses information dated after t. Construct a surrogate series by replacing x_t with $x_{t'}$ for t' drawn from $\mathcal{N}(t)$, recompute $\hat{I}(X;Y \mid Z)$ on the surrogate, repeat B times, and reject $H_0 : I(X;Y \mid Z) = 0$ if the observed statistic exceeds the $(1-\alpha)$ quantile of the surrogate distribution.

Restricting $\mathcal{N}(t)$ to the causal past mirrors the point-in-time discipline underlying `SignalOptimizer`'s use of sequential instead of random cross-validation.

Linear Benchmark and Nonlinearity Diagnostic

The estimator above reports total dependence, linear and nonlinear combined, without indicating how much of it a linear regression screen would already have found. Separating the two requires a benchmark computable under a purely linear model, against which the nonparametric estimate can be compared directly.

Proposition 2 (Gaussian conditional mutual information, e.g. Gaussian Graphical Models, 2011)**.** *For jointly Gaussian (X, Y, Z),*

$$I_{\text{Gauss}}(X;Y \mid Z) = -\tfrac{1}{2} \log \left(1 - \rho^2(X, Y \mid Z)\right),$$

where $\rho(X, Y \mid Z)$ is the partial correlation of X and Y given Z.

I_{Gauss} is the quantity a linear regression screen effectively measures. Define

$$\Delta(X) = \hat{I}_{k\text{NN}}(X; r_{t+h} \mid \mathcal{I}_t) - I_{\text{Gauss}}(X; r_{t+h} \mid \mathcal{I}_t).$$

$\Delta(X) \approx 0$ indicates linear, OLS-recoverable incremental information; $\Delta(X) \gg 0$ indicates nonlinear information a linear screen discards. I_{Gauss} also serves as a fallback where $\hat{I}_{k\text{NN}}$ is unreliable at a given d, since its accuracy does not depend on d in the same way.

Computational Tractability

Pooling addresses the sample-size side of the constraint above but not runtime. Estimating $I(X;Y \mid Z)$, with significance testing, separately for every candidate indicator against a large conditioning set is expensive regardless of how much data is available per estimate.

Procedure 1 (First-stage filtering, following the PC_1 step of Runge, 2019)**.** *For each candidate indicator X, compute an inexpensive unconditional dependence statistic (NMI or linear correlation) against r_{t+h} using only information available at t; retain the top M candidates by this statistic to form a reduced conditioning set; apply the full conditional test above only within this reduced set.*

This addresses the cost documented in Causal Forecasting [2024] (approximately two weeks for unpruned conditional testing over 136 predictors), which is prohibitive at JPMaQS scale.

Horizon Dependence

An indicator's incremental information need not be constant across the forecast horizon h. Some indicators are informative only in the very short run, while others remain informative for months. Practitioners commonly summarise this by tracking a signal's information coefficient, the correlation between the signal and forward returns, across increasing horizons, and reporting the horizon at which this correlation falls to half its peak value as the signal's half-life; this captures decay in a linear sense only.

Definition 6 (Information half-life)**.**

$$h^\star = \min\left\{ h : I(X_t; r_{t+h} \mid \mathcal{I}_t) \le \tfrac{1}{2} \max_{h'} I(X_t; r_{t+h'} \mid \mathcal{I}_t) \right\}.$$

I compute $h^\star$ alongside the analogous half-life of the linear information coefficient. Divergence between the two indicates the indicator's nonlinear content decays at a different rate than its linear content.

Empirical Plan

Compare the indicator set selected by a significance-tested conditional-mutual-information threshold to the set selected by `SignalOptimizer` under panel OLS; report disagreements together with each indicator's $\Delta(X)$ and $h^\star$. This project's novel comparison is whether nonlinear or higher-order dependence identified by conditional mutual information is systematically missed by the linear panel selection Macrosynergy's own tooling currently performs.

Request

I would like your assessment of whether this is well scoped for an undergraduate project, feedback on the pooling assumption and the estimator's dimensionality limits in particular, and, if the project seems worthwhile to you, your willingness to serve as the Princeton faculty sponsor for the JPMaQS data usage agreement with J.P. Morgan, which Macrosynergy's Program with Academia requires.

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